

Multi-Variable Shannon–Rényi Control Laws in Arithmetic Power Geometry: Theory, Empirical Validation, and Comparative Analysis on Cache-Like Information Networks

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Abstract

Arithmetic Power Geometry (APG) studies algebraic closure relations under exponent deformation by fixing positive base coordinates and treating the exponent as a continuous deformation parameter. Earlier APG work established that the two-coordinate local closure defect at the Euclidean target is controlled by Shannon and Rényi information structure. This paper develops the finite-dimensional generalization of that result. For a positive vector $X = (x_1, \dots, x_m)$, we define squared Euclidean weights $w_i = x_i^2 / \sum_{j=1}^m x_j^2$ and the normalized multi-variable APG defect $D_m(t) = 1 - \sum_{i=1}^m w_i^{t/2}$ for $t \geq 2$. We prove a bridge identity between the APG closure defect and the power-sum defect, establish a Shannon–Rényi upper bound

$$D_m(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right) \leq \frac{t-2}{2}H(W),$$

and prove a dimension-stable lower bound

$$D_m(t) \geq 1 - M^{(t-2)/2},$$

where $M = \max_i w_i$. The theoretical results are evaluated on MovieLens 100K treated as a cache-like request distribution with movies as objects and rating events as requests. The experiment confirms the APG bounds for $t \in \{2.5, 3, 3.5, 4, 5, 6\}$ and compares APG descriptors with Shannon entropy, Rényi entropy, Herfindahl–Hirschman concentration, Gini inequality, and Zipf popularity scaling. The results show that APG descriptors are mathematically valid and most informative at low deformation levels, especially $t = 2.5$ and $t = 3.0$, while larger exponents saturate rapidly. The paper positions APG not as a replacement for classical entropy or cache metrics, but as a bounded deformation descriptor for concentration growth in high-dimensional information systems.

Keywords: Arithmetic Power Geometry; Shannon entropy; Rényi entropy; cache workload; MovieLens; Zipf law; information geometry; concentration measure; power-sum defect.

1 Introduction

Large information systems often exhibit uneven access distributions: a small number of objects receive a substantial portion of requests, while a long tail of objects receives only sparse access. This pattern is central in web caching, content delivery, recommender systems, key-value caches, and edge computing. Classical studies of web caching reported Zipf-like request distributions and used popularity skew to explain cache hit behavior (Breslau et al., 1999; Zipf, 1949). Modern production systems continue to show strong workload diversity, time-to-live effects, size heterogeneity, and popularity concentration at industrial scale (Berg et al., 2020; Mao et al., 2024; Sultan et al., 2024; Yang et al., 2020).

At the same time, information theory provides a mature language for measuring dispersion, uncertainty, and concentration. Shannon entropy measures uncertainty in a probability distribution (Cover & Thomas, 2006; Shannon, 1948). Rényi entropy generalizes Shannon entropy through a one-parameter family of power-sum measures (Rényi, 1961; van Erven & Harremoës, 2014). Related measures such as Tsallis entropy, the Herfindahl–Hirschman index, Gini inequality, and Zipf exponents are widely used to quantify inequality, concentration, and heavy-tailed behavior (Gini, 1912; Jain et al., 1991; Piantadosi, 2014; Tsallis, 1988).

Arithmetic Power Geometry (APG) introduces a complementary viewpoint: it studies how closure relations deform when the exponent is varied continuously over fixed positive base coordinates. APG I established that the local closure defect near the Euclidean target is governed, to first order, by Shannon entropy (Akhtar, 2026a). A later Shannon–Rényi control paper showed that the two-coordinate APG closure defect can be converted into a normalized power-sum defect and controlled by Rényi monotonicity (Akhtar, 2026b). However, that result was formulated primarily in the two-coordinate setting.

This paper addresses the finite-dimensional extension. The goal is not to claim a new theorem independent of classical information theory. Rather, the contribution is to show that the APG defect has a dimension-stable Shannon–Rényi control structure once the Euclidean weights are generalized to an arbitrary finite probability vector. This gives APG a mathematically clean path from two-coordinate exponent deformation to high-dimensional information systems.

The paper makes four contributions. First, it defines the multi-variable APG defect on m -dimensional positive coordinates. Second, it proves that the defect is exactly a normalized power-sum defect. Third, it establishes upper and lower bounds using Shannon entropy, Rényi entropy, and the maximum coordinate weight. Fourth, it evaluates the descriptors on MovieLens 100K treated as a cache-like request distribution and compares APG against existing concentration metrics. Unlike Shannon entropy, Rényi entropy, HHI, or Gini inequality, the APG descriptor originates from exponent-deformation geometry rather than from information-theoretic axioms. The contribution of this paper is therefore not the introduction of a new entropy measure, but the identification of classical power-sum concentration quantities as Euclidean-target closure defects arising naturally from Arithmetic Power Geometry.

2 Related Work

2.1 Entropy and Power-Sum Measures

Shannon entropy is the foundational uncertainty measure for discrete probability distributions (Shannon, 1948). Rényi entropy extends Shannon entropy into a spectrum of order-sensitive measures, with monotonicity in the order parameter serving as a key analytical property (Csiszár & Körner, 2011; Rényi, 1961; van Erven & Harremoës, 2014). Tsallis entropy provides another non-additive generalization based on power sums (Tsallis, 1988). These measures are closely connected to convexity, majorization, and classical inequality theory (Dembo et al., 1991; Hardy et al., 1952; Marshall et al., 2011).

2.2 Caching, Popularity Skew, and Zipf Behavior

Cache performance depends heavily on request concentration. Zipf-like popularity has been repeatedly observed in web and content-access workloads (Breslau et al., 1999; Piantadosi, 2014; Zipf, 1949). Classical cache models connect object popularity to miss ratio behavior and replacement decisions (Che et al., 2002; Megiddo & Modha, 2003; O’Neil et al., 1993). Recent production studies of cache clusters at Twitter and Facebook/Meta demonstrate that real workloads are heterogeneous and cannot be fully described by a single idealized distribution (Berg et al., 2020; Yang et al., 2020). Newer work emphasizes TTL-aware sizing, cache freshness, edge computing, and learned cache management (Jazaeri et al., 2023; Mao et al., 2024; Shahout et al., 2024; Sultan et al., 2024).

2.3 APG and the Present Extension

APG I introduced the Euclidean target and the first-order entropy-controlled local closure defect (Akhtar, 2026a). The Shannon–Rényi APG control paper converted the two-coordinate closure defect into a power-sum defect and bounded it using Rényi entropy monotonicity (Akhtar, 2026b). The present paper uses only those APG papers that are directly required for the definitions and the Shannon–Rényi bridge. It does not rely on later APG height-coupling or Arakelov-projection manuscripts.

3 Multi-Variable APG Framework

Let

$$X = (x_1, x_2, \dots, x_m) \in \mathbb{R}_+^m$$

be a vector of positive base coordinates. The generalized Euclidean target norm is

$$\|X\|_2 = \left(\sum_{i=1}^m x_i^2 \right)^{1/2}.$$

For a real deformation exponent $t > 0$, define the multi-variable Euclidean-target closure defect

$$\delta_K(t) = \left| 1 - \frac{\sum_{i=1}^m x_i^t}{(\sum_{i=1}^m x_i^2)^{t/2}} \right|.$$

Define squared Euclidean weights

$$w_i = \frac{x_i^2}{\sum_{j=1}^m x_j^2}, \quad i = 1, \dots, m.$$

Then $w_i > 0$ and $\sum_{i=1}^m w_i = 1$, so

$$W = (w_1, \dots, w_m)$$

is a probability vector. The Shannon entropy and maximum coordinate weight are

$$H(W) = -\sum_{i=1}^m w_i \log w_i, \quad M = \max_{1 \leq i \leq m} w_i.$$

For $t \geq 2$, define the normalized multi-variable APG defect

$$D_m(t) = 1 - \sum_{i=1}^m w_i^{t/2}.$$

Lemma 1 (Generalized APG normalization bridge). *For every $X \in \mathbb{R}_+^m$ and every $t \geq 2$,*

$$\delta_K(t) = D_m(t).$$

Proof. For each i ,

$$w_i^{t/2} = \left(\frac{x_i^2}{\sum_{j=1}^m x_j^2} \right)^{t/2} = \frac{x_i^t}{\left(\sum_{j=1}^m x_j^2 \right)^{t/2}}.$$

Summing gives

$$\sum_{i=1}^m w_i^{t/2} = \frac{\sum_{i=1}^m x_i^t}{\left(\sum_{j=1}^m x_j^2 \right)^{t/2}}.$$

Since $t/2 \geq 1$ and $0 < w_i < 1$, $w_i^{t/2} \leq w_i$. Therefore

$$\sum_{i=1}^m w_i^{t/2} \leq \sum_{i=1}^m w_i = 1,$$

so the quantity inside the absolute value is nonnegative. Hence

$$\delta_K(t) = 1 - \frac{\sum_{i=1}^m x_i^t}{\left(\sum_{j=1}^m x_j^2 \right)^{t/2}} = 1 - \sum_{i=1}^m w_i^{t/2} = D_m(t).$$

□

4 Main Results

Theorem 1 (Generalized Shannon–Rényi upper control). *Let W be an m -dimensional probability vector generated from positive APG coordinates. For every $t \geq 2$,*

$$D_m(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right) \leq \frac{t-2}{2}H(W).$$

Proof. Let $q = t/2$. Then $q \geq 1$. For $q \neq 1$, Rényi entropy is

$$H_q(W) = \frac{1}{1-q} \log \left(\sum_{i=1}^m w_i^q \right).$$

Rényi entropy is non-increasing in the order parameter, so $H_q(W) \leq H_1(W) = H(W)$ for $q \geq 1$. Since $1 - q \leq 0$, multiplying by $1 - q$ reverses the inequality:

$$\log \left(\sum_{i=1}^m w_i^q \right) \geq (1 - q)H(W).$$

Exponentiating gives

$$\sum_{i=1}^m w_i^q \geq \exp(-(q-1)H(W)).$$

Therefore

$$1 - \sum_{i=1}^m w_i^q \leq 1 - \exp(-(q-1)H(W)).$$

Substituting $q = t/2$ gives the first bound. The second follows from $1 - e^{-x} \leq x$ for $x \geq 0$. $\square$

Theorem 2 (Dimension-stable maximum-weight lower control). *Let $M = \max_i w_i$. For every $t \geq 2$,*

$$D_m(t) \geq 1 - M^{(t-2)/2}.$$

Proof. Let $q = t/2 \geq 1$. Since $w_i \leq M$ and $q - 1 \geq 0$,

$$w_i^{q-1} \leq M^{q-1}.$$

Multiplying by $w_i > 0$ gives

$$w_i^q \leq w_i M^{q-1}.$$

Summing over i ,

$$\sum_{i=1}^m w_i^q \leq M^{q-1} \sum_{i=1}^m w_i = M^{q-1}.$$

Thus

$$1 - \sum_{i=1}^m w_i^q \geq 1 - M^{q-1}.$$

Substituting $q = t/2$ proves the result. $\square$

5 Experimental Evaluation

5.1 Dataset

The empirical test uses MovieLens 100K, a widely used benchmark dataset containing 100,000 ratings (Harper & Konstan, 2015). In this study, the dataset is treated as a cache-like request distribution in which each movie is regarded as an object and each rating event is interpreted as a request for that object.

5.2 Metrics

Let r_i denote the number of requests associated with object i , and define

$$w_i = \frac{r_i}{\sum_{j=1}^m r_j}.$$

The following descriptors were computed:

$$H(W) = - \sum_i w_i \log w_i,$$

$$H_q(W) = \frac{1}{1-q} \log \left(\sum_i w_i^q \right),$$

$$\text{HHI} = \sum_i w_i^2,$$

and

$$D_m(t) = 1 - \sum_i w_i^{t/2}.$$

5.3 Dataset Summary

Table 1 reports the basic characteristics of the MovieLens 100K request distribution.

Table 1: MovieLens 100K cache-like distribution summary.

Quantity	Value
Request events	100,000
Objects/movies	1,682
Shannon entropy $H(W)$	6.7279
Normalized entropy $H(W)/\log m$	0.9058
Maximum weight M	0.00583
HHI	0.00168
Gini coefficient	0.6290
Zipf exponent estimate α	1.3955
Log-log Zipf R^2	0.7182

Figure 1 shows the long-tail popularity structure of the MovieLens 100K distribution. A small number of objects receive a large fraction of requests, while a long tail of objects receives relatively few requests. This behavior is consistent with the observed Gini coefficient and Zipf exponent reported in Table 1.

Table 2 presents a quantitative comparison between APG and existing concentration measures.

The values reported in Table 2 indicate that the MovieLens distribution combines high diversity with substantial popularity concentration.

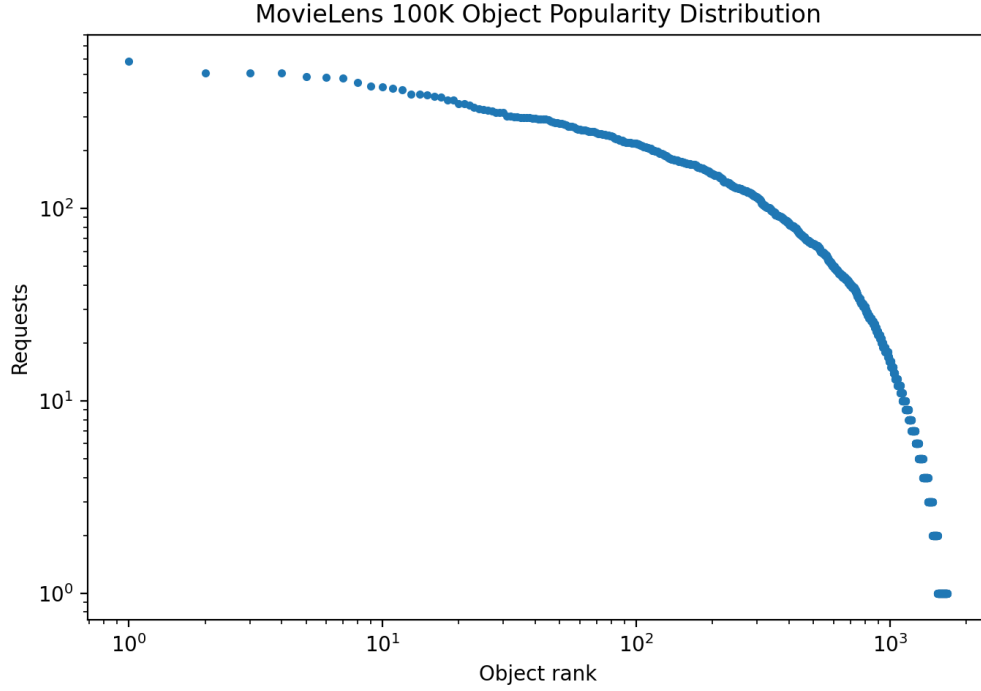


Figure 1: Rank-frequency popularity distribution of MovieLens 100K objects.

Table 2: Quantitative comparison of APG with existing concentration measures on MovieLens 100K.

Metric	Value
Shannon entropy $H(W)$	6.7279
Renyi entropy $H_{1.25}(W)$	6.41
Herfindahl–Hirschman Index (HHI)	0.00168
Gini coefficient	0.6290
Zipf exponent α	1.3955
APG defect $D_m(2.5)$	0.8091
APG defect $D_m(3.0)$	0.9619

Table 3: Verification of APG lower and upper bounds on MovieLens 100K.

t	Lower bound	$D_m(t)$	Upper bound	Valid
2.5	0.7237	0.8091	0.8140	Yes
3.0	0.9236	0.9619	0.9654	Yes
3.5	0.9789	0.9921	0.9936	Yes
4.0	0.9942	0.9983	0.9988	Yes
5.0	0.9996	0.9999	1.0000	Yes
6.0	1.0000	1.0000	1.0000	Yes

5.4 APG Bound Verification

Table 3 reports the lower bound, APG defect, and upper bound for several deformation exponents.

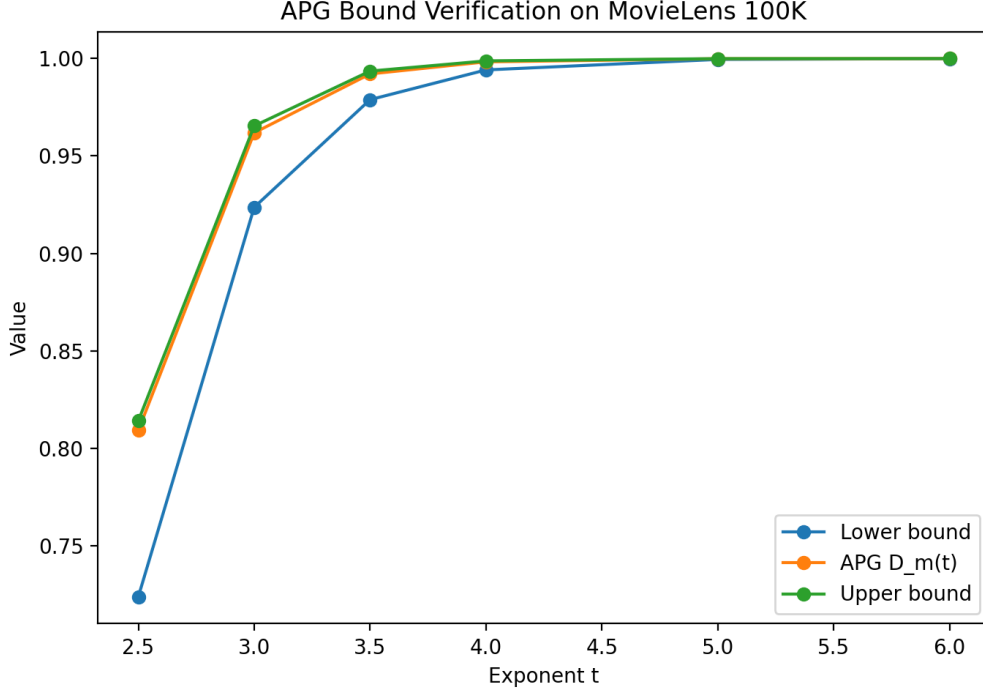


Figure 2: APG lower bound, APG defect, and Shannon–Rényi upper bound on MovieLens 100K.

Figure 2 visually confirms the theoretical result. For every tested deformation exponent, the APG defect remains between the derived lower and upper bounds, with the gap narrowing as t increases.

As shown in Table 3, the theoretical inequalities are satisfied for all tested deformation levels. The most informative APG settings are $t = 2.5$ and $t = 3.0$. For $t \geq 4$, the APG defect approaches unity.

5.5 Comparison With Existing Descriptors

Table 4 summarizes the conceptual interpretation of the descriptors considered in this study.

The quantitative results in Table 2 complement the qualitative interpretation in Table 4.

5.6 Relationship Between APG and HHI

An important special case occurs at $t = 4$. Since

$$D_m(t) = 1 - \sum_{i=1}^m w_i^{t/2},$$

setting $t = 4$ yields

$$D_m(4) = 1 - \sum_{i=1}^m w_i^2.$$

Table 4: Comparison of APG with standard information and concentration descriptors.

Descriptor	Interpretation	Result on MovieLens 100K
Shannon entropy	Overall uncertainty/diversity	High normalized diversity 0.9058
Renyi entropy	Order-sensitive power-sum diversity	Decreases as q increases
HHI	Quadratic concentration	Low global concentration 0.00168
Gini coefficient	Inequality of object requests	Moderate-high skew 0.6290
Zipf exponent	Rank-frequency decay	Heavy-tail behavior, $\alpha = 1.3955$
APG defect $D_m(t)$	Exponent-deformation concentration defect	Most informative at $t = 2.5$ and $t = 3.0$

Because the Herfindahl–Hirschman Index is defined as

$$\text{HHI} = \sum_{i=1}^m w_i^2,$$

we obtain

$$D_m(4) = 1 - \text{HHI}.$$

Thus, HHI appears as a special point within the APG deformation spectrum.

6 Discussion

The experiment supports the theoretical claim that the APG defect is bounded by Shannon–Rényi structure in finite dimensions. However, APG should not be overstated as an operational replacement for Shannon entropy, Rényi entropy, HHI, Gini coefficient, or established cache models. Its main value is interpretive: it describes how concentration changes under exponent deformation and gives certified upper and lower limits.

For cache-like systems, APG may be useful as a supplementary descriptor for concentration growth, hotspot formation, or anomaly monitoring. Production cache studies emphasize that real systems involve TTLs, object sizes, write patterns, and workload heterogeneity (Berg et al., 2020; Mao et al., 2024; Sultan et al., 2024; Yang et al., 2020). The present MovieLens experiment does not include those operational variables. Therefore, CDN claims should be framed cautiously until validated on true cache traces such as Twitter production cache traces or CacheLib-style workloads.

7 Limitations

This study has five limitations. First, MovieLens 100K is a recommendation dataset, not a true CDN trace. Second, the APG defect saturates rapidly for larger t , so low deformation exponents should be preferred for empirical discrimination. Third, the comparison is descriptor-level rather than a full cache-policy benchmark against LRU, LFU, ARC, TinyLFU, or learned cache policies. Fourth, the main theorem is a structured APG

translation of classical Rényi monotonicity, not an independent replacement for established information theory. Fifth, the present study demonstrates mathematical validity rather than predictive superiority. Additional experiments on fraud detection, anomaly detection, blockchain transaction networks, and production cache traces are required to determine whether APG descriptors provide practical advantages over existing concentration measures.

8 Conclusion

This paper generalized the Shannon–Rényi APG control law from the two-coordinate setting to arbitrary finite dimension. The multi-variable APG closure defect was shown to be exactly equivalent to a normalized power-sum defect, allowing the use of classical information-theoretic tools to study exponent-deformation behavior in APG. Using Rényi entropy monotonicity, we established a finite-dimensional Shannon–Rényi upper bound and a complementary maximum-weight lower bound for the APG defect.

Empirical evaluation on the MovieLens 100K dataset confirmed the theoretical bounds across all tested deformation levels. The results further showed that lower deformation exponents, particularly $t = 2.5$ and $t = 3.0$, provide the greatest discriminatory power, whereas larger exponents rapidly approach saturation. An additional observation is that the APG deformation spectrum naturally contains classical concentration measures as special cases. In particular,

$$D_m(4) = 1 - \text{HHI},$$

showing that the Herfindahl–Hirschman Index can be viewed as a specific point within the broader APG framework.

The principal contribution of this work is the identification of a finite-dimensional bridge between exponent-deformation geometry and classical information concentration measures. Rather than introducing a new entropy measure, the paper provides a geometric interpretation of power-sum concentration through Euclidean-target closure defects. In this sense, APG complements existing entropy-based approaches by supplying an exponent-deformation perspective on concentration phenomena.

The present results establish mathematical validity and finite-dimensional stability, but they do not by themselves demonstrate superiority over existing concentration descriptors. Future work should therefore evaluate APG on fraud detection, anomaly detection, blockchain transaction networks, and production-scale cache traces to determine whether exponent-deformation descriptors provide practical advantages in real-world settings.

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